

3-Mark Questions & Answers — MPCA Module 4: Memory & Peripherals Interfacing

Q1. Explain the Control Word Register (CWR) format of 8255 PPI in I/O mode.

- **Bit D7 = 1:** I/O Mode Set flag.
- **Bits D6-D5:** Group A Mode (00=Mode 0, 01=Mode 1, 1X=Mode 2).
- **Bit D4:** Port A (1=Input, 0=Output).
- **Bit D3:** Port C Upper (1=Input, 0=Output).
- **Bit D2:** Group B Mode (0=Mode 0, 1=Mode 1).
- **Bit D1:** Port B (1=Input, 0=Output).
- **Bit D0:** Port C Lower (1=Input, 0=Output).

Q2. Explain the DMA transfer cycle between 8257 and 8086 CPU.

1. Peripheral asserts **DRQ** to 8257.
2. 8257 asserts **HRQ** to 8086 CPU.
3. 8086 floats system buses and asserts **HLDA** to 8257.
4. 8257 asserts **DACK**, puts memory address on bus, and executes single-cycle I/O to memory transfer.
5. On reaching Terminal Count (**TC**), 8257 drops **HRQ**, releasing buses to CPU.

Q3. Explain 8259 cascading in Master-Slave mode.

- One **Master 8259** connects to up to **8 Slave 8259s** via the 3-line Cascade Bus (**CAS0**, **CAS1**, **CAS2**).
- Increases interrupt handling capacity from 8 to **64 priority channels**.